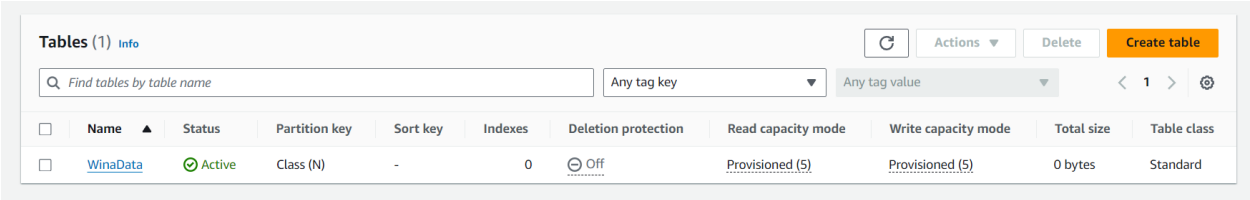
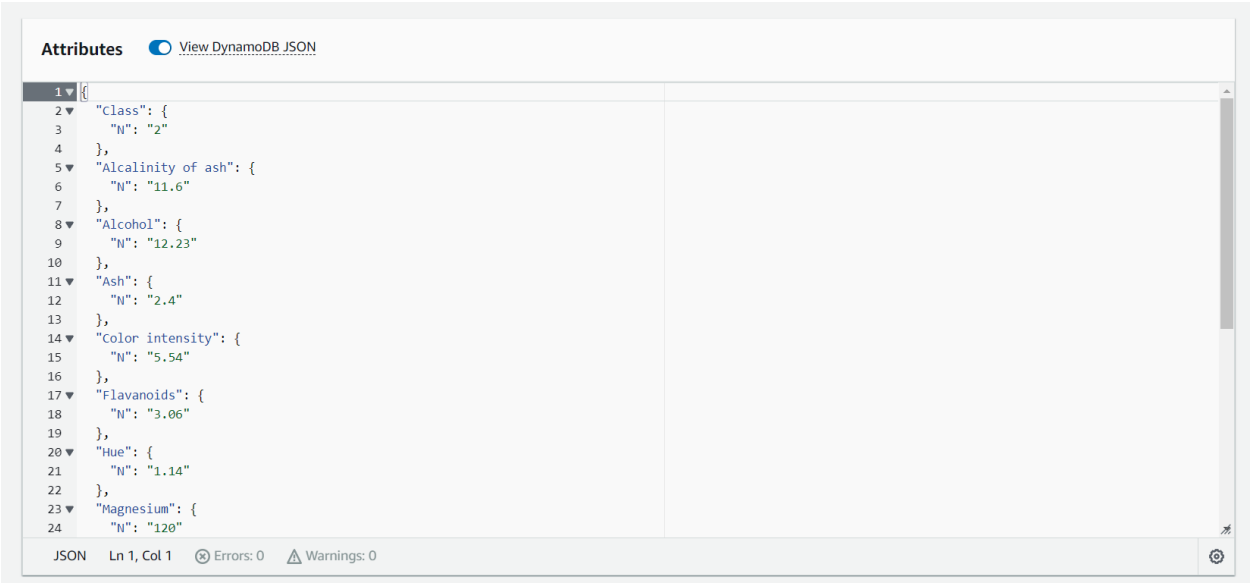
Створюємо таблицю:

```
┌──(smth㉿smth)-[~]
└─$ aws dynamodb create-table --table-name WinaData --attribute-definitions AttributeName=Class,AttributeType=N --key
-schema AttributeName=Class,KeyType=HASH --provisioned-throughput ReadCapacityUnits=5,WriteCapacityUnits=5
{
    "TableDescription": {
        "AttributeDefinitions": [
            {
                "AttributeName": "Class",
                "AttributeType": "N"
            }
        ],
        "TableName": "WinaData",
        "KeySchema": [
            {
                "AttributeName": "Class",
                "KeyType": "HASH"
            }
        ],
        "TableStatus": "CREATING",
        "CreationDateTime": "2024-06-29T00:25:43.842000-04:00",
        "ProvisionedThroughput": {
            "NumberOfDecreasesToday": 0,
            "ReadCapacityUnits": 5,
            "WriteCapacityUnits": 5
        },
        "TableSizeBytes": 0,
        "ItemCount": 0,
        "TableArn": "arn:aws:dynamodb:eu-north-1:891377288524:table/WinaData",
        "TableId": "fd20b846-8837-4134-a54e-dd8ed0dcc39d",
        "DeletionProtectionEnabled": false
    }
}
```

Бачимо її у AWS:



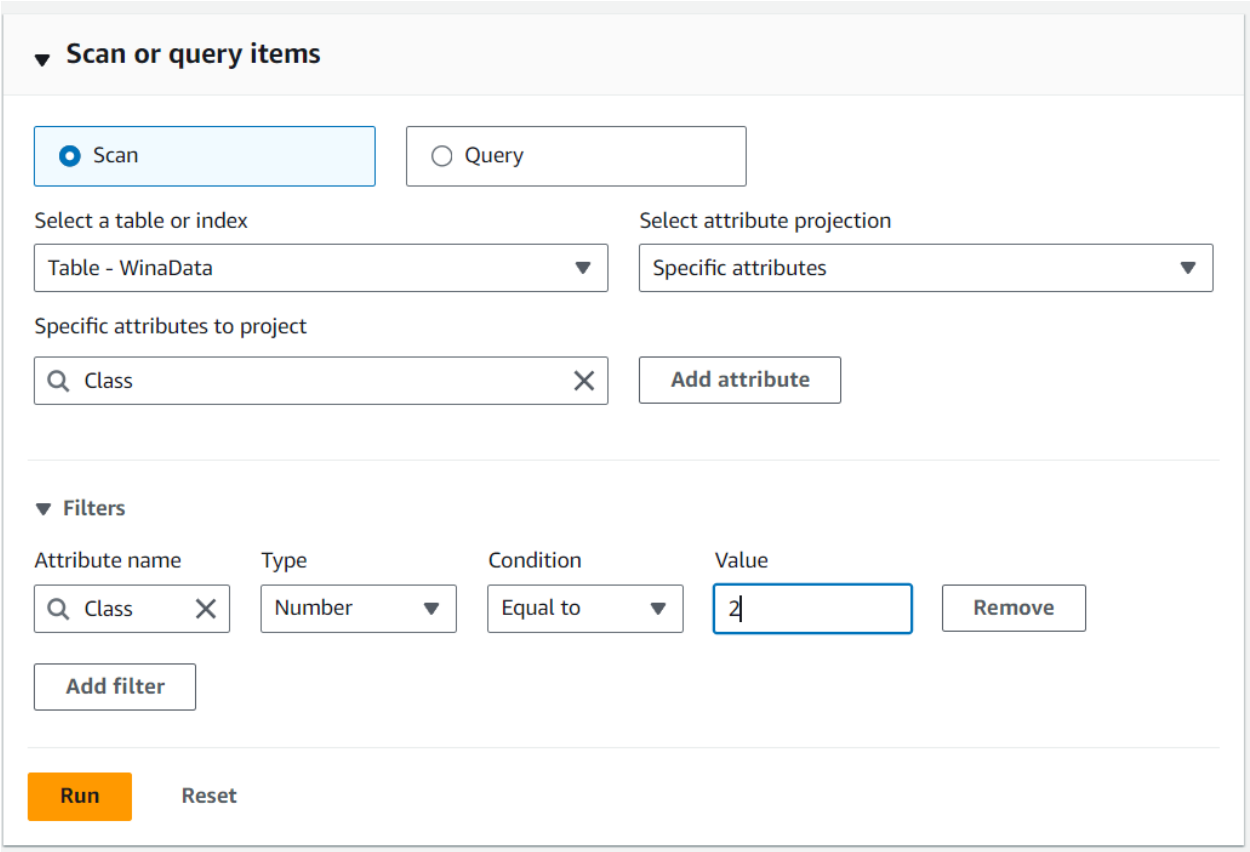
Створюємо по семплу в AWS CLI та AWS Management Console:



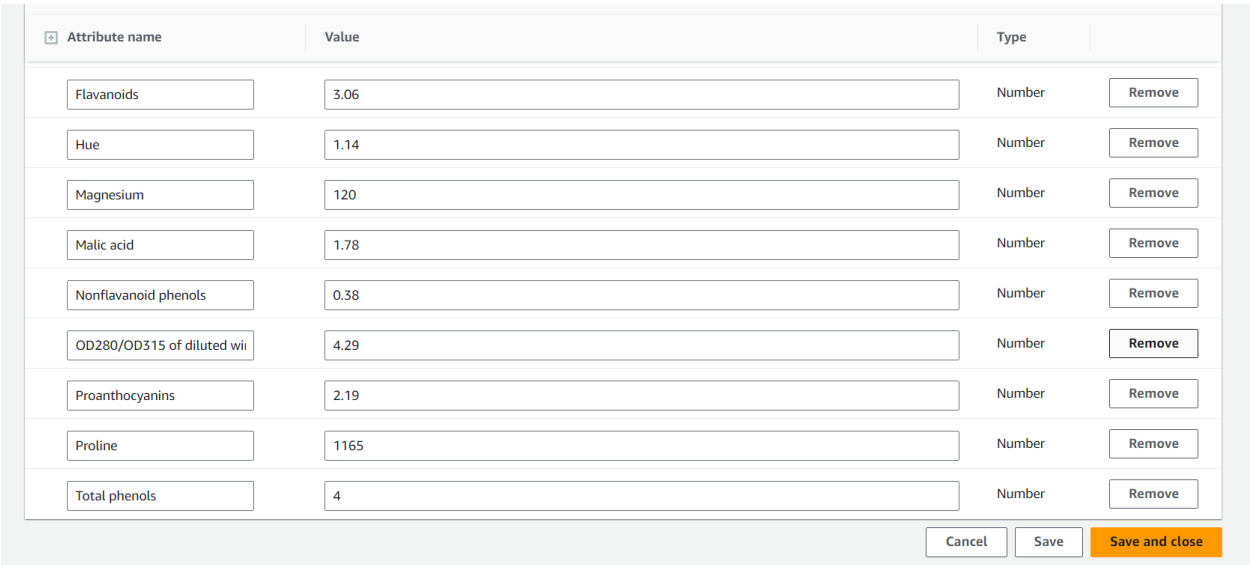
```
┌──(smth㉿smth)-[~]
└─$ aws dynamodb put-item --table-name WinaData --item '{"Class": {"N": "1"},
        "Alcohol": {"N": "14.23"},
        "Malic acid": {"N": "1.71"},
        "Ash": {"N": "2.43"},
        "Alcalinity of ash": {"N": "15.6"},
        "Magnesium": {"N": "127"},
        "Total phenols": {"N": "2.8"},
        "Flavanoids": {"N": "3.06"},
        "Nonflavanoid phenols": {"N": "0.28"},
        "Proanthocyanins": {"N": "2.29"},
        "Color intensity": {"N": "5.64"},
        "Hue": {"N": "1.04"},
        "OD280/OD315 of diluted wines": {"N": "3.92"},
        "Proline": {"N": "1065"}}'
```

Пошук даних:

```
┌──(smth㉿smth)-[~]
└─$ aws dynamodb get-item --table-name WinaData --key '{"Class": {"N": "1"}}'
{
    "Item": {
        "Class": {
            "N": "1"
        },
        "Flavanoids": {
            "N": "3.06"
        },
        "Alcohol": {
            "N": "14.23"
        },
        "Proanthocyanins": {
            "N": "2.29"
        },
        "Total phenols": {
            "N": "2.8"
        },
        "Color intensity": {
            "N": "5.64"
        },
        "Proline": {
            "N": "1065"
        },
        "Nonflavanoid phenols": {
            "N": "0.28"
        },
        "OD280/OD315 of diluted wines": {
            "N": "3.92"
        },
        "Alcalinity of ash": {
            "N": "15.6"
        },
        "Ash": {
            "N": "2.43"
        },
        "Hue": {
            "N": "1.04"
        }
```



Їх модифікація:




```
┌──(smth㉿smth)-[~]
└─$ aws dynamodb update-item --table-name WinaData --key '{"Class": {"N": "1"}}' --update-expression "set Alcohol = :
a" --expression-attribute-values '{":a": {"N": "17"}}'
```

Та видалення:

```
┌──(smth㉿smth)-[~]
└─$ aws dynamodb delete-item --table-name WinaData --key '{"Class": {"N": "2"}}'
```

```python
import boto3
from botocore.exceptions import NoCredentialsError, PartialCredentialsError
import json

dynamodb = boto3.resource('dynamodb', region_name='eu-north-1')

def create_table():
    try:
        table = dynamodb.create_table(
            TableName='WineData',
            KeySchema=[
                {
                    'AttributeName': 'Class',
                    'KeyType': 'HASH'
                },
                {
                    'AttributeName': 'ID',
                    'KeyType': 'RANGE'
                }
            ],
            AttributeDefinitions=[
                {
                    'AttributeName': 'Class',
                    'AttributeType': 'N'
                },
                {
                    'AttributeName': 'ID',
                    'AttributeType': 'N'
                }
            ],
            ProvisionedThroughput={
                'ReadCapacityUnits': 5,
                'WriteCapacityUnits': 5
            }
        )
        table.wait_until_exists()
        print("Table created successfully.")
    except Exception as e:
        print(f"Error creating table: {e}")

def insert_data():
    table = dynamodb.Table('WineData')
    try:
        table.put_item(
            Item={
                'Class': 1,
                'ID': 1,
                'Alcohol': 14.23,
                'Malic acid': 1.71
            }
        )
        table.put_item(
            Item={
                'Class': 1,
                'ID': 2,
                'Alcohol': 13.20,
                'Malic acid': 1.78
            }
        )
        print("Data inserted successfully.")
    except Exception as e:
        print(f"Error inserting data: {e}")

def read_data():
    table = dynamodb.Table('WineData')
```

```
    try:
        response = table.get_item(
            Key={
                'Class': 1,
                'ID': 1
            }
        )
        item = response.get('Item')
        print(item)
    except Exception as e:
        print(f"Error reading data: {e}")

def update_data():
    table = dynamodb.Table('WineData')
    try:
        table.update_item(
            Key={
                'Class': 1,
                'ID': 1
            },
            UpdateExpression='SET Alcohol = :val1',
            ExpressionAttributeValues={
                ':val1': 15.0
            }
        )
        print("Data updated successfully.")
    except Exception as e:
        print(f"Error updating data: {e}")

def delete_data():
    table = dynamodb.Table('WineData')
    try:
        table.delete_item(
            Key={
                'Class': 1,
                'ID': 1
            }
        )
        print("Data deleted successfully.")
    except Exception as e:
        print(f"Error deleting data: {e}")

def load_json_data(file_name):
    table = dynamodb.Table('WineData')
    try:
        with open(file_name) as json_file:
            wines = json.load(json_file)
            for wine in wines:
                table.put_item(Item=wine)
        print("Data loaded from JSON file successfully.")
    except Exception as e:
        print(f"Error loading data from JSON file: {e}")


def query_data():
    table = dynamodb.Table('WineData')
    try:
        response = table.query(
            KeyConditionExpression=boto3.dynamodb.conditions.Key('Class').eq(1)
        )
        items = response.get('Items')
        for item in items:
            print(item)
    except Exception as e:
        print(f"Error querying data: {e}")
```

```python
def scan_data():
    table = dynamodb.Table('WineData')
    try:
        response = table.scan()
        items = response.get('Items')
        for item in items:
            print(item)
    except Exception as e:
        print(f"Error scanning table: {e}")

def delete_table():
    try:
        table = dynamodb.Table('WineData')
        table.delete()
        table.wait_until_not_exists()
        print("Table deleted successfully.")
    except Exception as e:
        print(f"Error deleting table: {e}")


create_table()
insert_data()
read_data()
update_data()
delete_data()
load_json_data('/home/smth/Downloads/relaxed.json')
query_data()
scan_data()
delete_table()
```